

The Mathematics of Money

Chapter 2: Present Value, Future Value, and Loan Amortization

Study Notes Series

1 Discounting: Running Compound Interest Backwards

Chapter 1 asked how much a present sum grows into over time. This chapter asks the reverse question: given a future amount you want or expect to receive, what is it worth today? This is called **present value (PV)**, and the process of computing it is called **discounting**.

If FV is a future value received in n years and r is the discount rate (annual, compounded once per year):

$$PV = \frac{FV}{(1+r)^n}. \quad (1)$$

The intuition: money in the future is worth less than money today, because today's money could have been invested at rate r and grown. A higher discount rate or a longer time horizon both shrink present value.

Worked example

What is the present value of \$10,000 to be received in 15 years, using a discount rate of $r = 4\%$?

$$PV = \frac{10,000}{(1.04)^{15}} \approx \frac{10,000}{1.8009} \approx \$5552.68. \quad (2)$$

2 Annuities: A Stream of Equal Payments

Many financial products (retirement savings, loans, bonds) involve a series of equal payments PMT made at regular intervals rather than a single lump sum. The future value of an ordinary annuity, where payments are made at the end of each period for n periods at rate r , is:

$$FV_{\text{annuity}} = PMT \times \frac{(1+r)^n - 1}{r}. \quad (3)$$

The corresponding present value of that same stream of payments is:

$$PV_{\text{annuity}} = PMT \times \frac{1 - (1+r)^{-n}}{r}. \quad (4)$$

3 Loan Amortization

Turning the annuity formula around gives the standard formula for a fixed loan payment, used for mortgages and car loans. If P is the loan principal, r is the periodic interest rate, and n is the number of payment periods:

$$PMT = P \times \frac{r(1+r)^n}{(1+r)^n - 1}. \quad (5)$$

Worked example: a simplified mortgage

Consider a \$200,000 loan at a monthly interest rate of $r = 0.005$ (roughly 6% annual, compounded monthly) over $n = 360$ months (30 years):

$$PMT = 200,000 \times \frac{0.005(1.005)^{360}}{(1.005)^{360} - 1} \approx \$1199.10 \text{ per month.} \quad (6)$$

Over the full 30 years, total payments amount to $360 \times 1199.10 \approx \$431,676$, meaning roughly \$231,676 of that is interest, more than the original principal. This illustrates why the interest rate and loan term matter enormously: shortening the term or lowering the rate can save tens of thousands of dollars over the life of a loan.

Quick check

Does increasing the loan term n while holding P and r fixed increase or decrease the monthly payment PMT ? What happens to total interest paid? (Answer: monthly payment decreases, but total interest paid increases, since interest accrues over more periods.)